The unreasonable effectiveness of mathematics, solved: mathematics as a substrate-physical bootstrap from a single root

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Wigner's 1960 observation that mathematics is "unreasonably effective" in describing nature rests on a tacit assumption: that mathematics is invented in pure abstraction and reality is a separate object onto which it is later mapped, with the success of the mapping requiring explanation. We propose that the puzzle dissolves when the assumption is dropped. In a substrate-physical framework in which a single ontological root — *the substrate exists* — is taken as the foundational hypothesis, every primitive of *the mathematics that physics uses* admits a substrate-physical reading: counting from event-counting, the natural numbers from extension-by-coupling-event, induction as a structural consequence of the substrate's iteration primitives, addition from linear superposition of substrate amplitudes in the free-propagation regime, multiplication capturing the substrate's non-linear cross-term coupling in its scalar commutative special case, division as the formal inverse with binding-energy substrate tension as the unmodelled remainder, the integers from opposing-tally processes, the rationals from observed ratios of counted quantities, the reals as Cauchy-completion against measurement, sets from substrate-type formation, topologies from measurement-precision neighbourhoods, smooth manifolds from elastic local flatness, differential forms from oriented coupling events, Lie groups from continuous symmetries, Lie algebras from infinitesimal symmetry generators, root systems from quantised standing-wave charges, and Coxeter groups from the substrate's reflection structure on its mode lattice. Each construction is paired with an observable discharge whose existence in laboratory or astrophysical data licenses the construction. On this reading, mathematics is not a separate language that happens to fit physics; it is the formal description of the substrate's primitive content, and its effectiveness in physics is not a coincidence to be explained but a structural consequence of the identity between description and described. We present the bootstrap chain end-to-end, identify the empirical anchor that the framework's E_8 root system shares with the observed Zamolodchikov spectrum in $CoNb_2O_6$, and discuss the open work of fully discharging the chain.

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I. INTRODUCTION

In his 1960 essay [1], Eugene Wigner asked why a body of abstract mathematical structure — complex numbers, Hilbert spaces, group representations, differential geometry — developed without reference to physics should turn out to describe physics so accurately that "it is difficult to

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avoid the impression that a miracle is involved." Two of his examples are paradigmatic. Heisenberg's matrix mechanics succeeded in spite of the fact that "the rules of matrix mechanics had been derived from problems which involved the old theory of the hydrogen atom"; complex numbers, used at the time mostly as a calculational convenience for real wavefunctions, turned out to be irreducible to quantum mechanics. The puzzle Wigner left us is sharp: how does a body of structure invented in pure abstraction fit a body of phenomena measured in experiments, when the two were not constructed in concert?

Wigner's question is usually treated in one of three ways. The *platonist* reply [2] grants mathematics independent reality and treats its match with physics as an observation about that reality. The *naturalist* reply [3, 4] treats mathematics as a refined extract from physical practice and argues the match is what one would expect. The *deflationary* reply [5, 6] questions whether the success is unreasonable at all, observing that the parts of mathematics most useful to physics tend to be the parts physics asked for. Each of these positions has its merits and its costs. None of them identifies a single technical commitment whose adoption would make the success of mathematics in physics a strict theorem rather than an unexplained tendency.

This paper proposes such a commitment. In a substrate-physical framework in which the foundational hypothesis is that *the substrate exists* — that there is a single ontological commitment to a coupled, event-counting, type-supporting substrate from which all observed phenomena unfold — we show that each primitive of the mathematics that physics uses is derivable from the substrate's primitive content, discharged by an observable phenomenon, and identical to the corresponding standard-mathematics object up to a bridge theorem (Sec. II A restricts the scope and lists what is *not* being claimed). The chain runs from the natural numbers through the rationals, the reals, sets, topologies, smooth manifolds, tensor calculus, differential forms, Lie groups, Lie algebras, root systems, and Coxeter groups; each is anchored in a specific substrate-physical process, and each carries an empirical signature that has already been observed.

On this reading, the answer to Wigner's question is structural and short. Mathematics is not a separate body of structure that happens to fit physics. The mathematics that appears in physics is, by construction, the formal description of the substrate's primitive content, and the success of that description on physical phenomena is a structural identity rather than a discovered fit. The "unreasonable effectiveness" of mathematics in the natural sciences ceases to be unreasonable when the mathematics-of-physics and the natural sciences are recognised as descriptions of a common source *at different levels of formalisation*. We do not extend this claim to mathematics that has no observed physical instantiation (see Sec. II A).

The paper is organised as follows. Section II introduces the framework's foundational ontology, identifying the single root commitment from which the bootstrap proceeds. Section III traces the chain from existence to the natural numbers and arithmetic, then shows that the four arithmetic operations have a parallel substrate-amplitude reading at the wave-pattern level. Section IV extends the chain to the integers, rationals, reals, sets, and topologies. Section V extends the chain through smooth manifolds, tensors, and differential forms. Section VI continues into Lie groups, Lie algebras, root systems, and Coxeter groups, and identifies the empirical anchor in $CoNb_2O_6$ that distinguishes the substrate's specific symmetry algebra from a generic one. Section VII states the resolution of Wigner's puzzle as a structural identity rather than a coincidence. Section VIII flags the limits of the present construction — which steps are observationally discharged, which are provisional, and what remains to be done.

II. THE FRAMEWORK'S FOUNDATIONAL ONTOLOGY

A. What this paper does and does not claim

Before stating the substrate's primitive content, we fix the scope of the argument, because several adjacent claims that the paper does *not* make would, if attributed to it, defeat it immediately.

a. What "substrate" means here. We use *substrate* as the framework's primitive ontological term for the underlying coupled, event-counting medium from which observed phenomena unfold. The substrate is not given a formal mathematical definition in this paper. Such a definition would be circular within a paper deriving mathematics from substrate primitives; the substrate is the ontological commitment whose primitive content the paper's bootstrap describes formally. The reader who insists on a mathematical characterisation may treat "substrate" as a placeholder for "the empirical object whose properties are listed below" and read the paper as a structural-equivalence claim between that list and a specific bootstrap of standard mathematics.

b. What "single root" means here. The framework's implication graph terminates, transitively, in a single ontological commitment: that the substrate exists. It does not terminate in a single *rule*. The substrate's primitive content (Sec. II B) is rich — it includes classical inference, type formation, And/Or/Not-coupling, Test, Loop, extensionality, and infinity. Calling this single-rooted is a statement about the implication graph, not a claim that the substrate's content is a single primitive operation. The paper does not claim mathematics is bootstrapped from *nothing*; it is bootstrapped from the substrate's primitive content, and that content is what makes the bootstrap go.

c. The hermeneutic circle. The bootstrap below uses higher-order logic, predicates, and quantification to describe substrate properties from which higher-order logic, predicates, and quantification are then said to fall out. This is the standard circularity of foundational projects (cf. Russell– Whitehead's use of logic to formalise logic, Lawvere's use of category theory to axiomatise category theory, the type-theoretic bootstrap of HoTT). It is not vicious, but it is real, and we note it explicitly: the paper's claim is that the substrate's primitive content is *coextensive* with what we call the metalogic, not that the metalogic is derived from a smaller starting point.

d. Distinction from the Mathematical Universe Hypothesis. The claim here is not Tegmark's Mathematical Universe Hypothesis [6]. The MUH says the universe *is* a mathematical structure, with all mathematical structures equally real. This paper's claim is narrower and asymmetric: the substrate has empirical content (specific particle masses, a specific symmetry algebra, specific topology) that selects one configuration out of the broader space of mathematical possibilities, and mathematics is what the substrate's primitive content looks like when described formally. The substrate is a particular thing; mathematics is its formal description; not every mathematical structure is a substrate, and the substrate's specific identity is empirical.

e. Mathematics beyond physics. We do not claim every mathematical object has a substrate-physical referent. Large cardinals, exotic forcing extensions, certain Banach–Tarski-type constructions, and other consequences of non-constructive choice principles have no observed substrate-physical instantiation. The paper's positive claim is restricted: each primitive of *the mathematics that physics uses* — elementary number theory, real analysis, point-set topology, smooth manifold theory, tensor calculus, exterior calculus, Lie theory, and the Coxeter classification — has a substrate-physical derivation and an observational discharge. Mathematics beyond that scope is neither denied nor explained here; it is consistent with the present construction that broader mathematics is generated by the closure of the substrate-derivable primitives under the framework's metalogic (extending well past anything observable), but we do not argue that here.

f. Bridge theorems. Each substrate-derived object below has both an existence-form definition (anchored in observation) and a constructive form (mirroring standard mathematics). The two presentations are connected by bridge theorems showing they describe the same object. The bridge theorems are stated and machine-verified throughout the framework's formal companion [11]; the present paper takes them as given and treats their full reproduction as out of scope. This is a load-bearing dependency, and we flag it as such.

g. Gödel-incompleteness. The substrate's primitive content (Sec. II B) suffices to interpret first-order Peano arithmetic, so any formal theory built from it inherits Gödel's first-incompleteness theorem: there exist arithmetical statements true in the standard model but not provable from the substrate-bootstrapped axioms. We do not treat this as a defect of the framework or as evidence against it. The incompleteness phenomenon is a property of any formal system rich enough to encode arithmetic, and the framework's substrate-physical discharge programme operates one level below the formal theory — it discharges *primitive content*, not the closure of all provable statements over that content. The Gödel sentence's substrate-physical status (whether it has any observable instance or is a purely logical artefact of the formalisation) is left as open metalogical work.

B. The substrate's primitive content

The framework's minimal ontology is committed to a single ontological root: the *substrate* exists. The substrate is a coupling-event-counting medium with the following primitive content, each piece of which is either grounded in the substrate's coupling-event definition or discharged by direct observation. We list the primitives because they are what the bootstrap will consume.

1. *Inference structure.* The substrate supports the standard logical inference rules: reflexivity, transitivity, application and abstraction congruence, beta reduction, modus ponens, hypothetical reasoning, antisymmetric deduction, variable instantiation, and type instantiation. Each rule is grounded in the substrate's coupling-event definition (a coupling event takes a state, applies an operation, resolves to a new state subject to admission criteria) and discharged by direct observation (chained-calibration agreement, biconditional substrate phenomena, exclusion-rule firing patterns).

2. *Foundational structure.* The substrate supports extensionality (identity-by-effect), eta-equivalence, selection from possibilities, the primitive operations Test, Loop, and And/Or/Not-coupling, and infinity in the sense that no finite bound is observed on the depth of recursive coupling.

3. *Bootstrap structure.* The substrate's event-counting satisfies two specific properties: *injectivity*, that extending two distinct event sequences each by one coupling event preserves their distinctness, and *non-zero successor*, that any sequence with at least one resolved event is distinguishable from the empty sequence. The substrate also supports stable *constant naming* (specific configurations retain identity across time and reference) and *type formation* (collections of configurations with a shared defining property are referred to as a kind).

These commitments are not framework conveniences. Each is a property of the universe that specific observ-

ables directly require: detector clicks have distinguishable counts, propagated light is distinguishable from unpropagated light, physical constants are constants, and electrons reproducibly share a single charge magnitude and mass.

We make explicit the commitment that does the work: every formal hypothesis the framework adopts carries an `impliedfrom` annotation listing the parent hypotheses it descends from, and a `discharge` annotation specifying the observational obligation that confirms it. Following the `impliedfrom` annotations transitively, every substrate hypothesis the framework commits to traces back to the single root: *the substrate exists*. The implication graph is, by construction, complete; what remains is the discharge work, which is empirical and collectively in progress [11].

III. FROM EXISTENCE TO ARITHMETIC

A. Counting and the natural numbers

The natural numbers are what the substrate produces by counting events. They start with zero — the count of the empty event-sequence, observable as the initial state of every counted process before any event has happened. The substrate then takes any count and produces the next count by applying its successor operation: one more coupling event has happened. Every detector click that goes from m clicks to $m+1$ clicks is the substrate doing this.

The natural numbers themselves are what one obtains by taking zero and applying successor again and again, without bound. The substrate's Loop construct iterates the successor application, and the substrate's infinity property guarantees the iteration never has to stop. The cumulative count of any counted substrate process — detector clicks, pulsar pulses, atomic-clock cycles — is a natural number.

Once the natural numbers exist, induction follows — not as an empirical generalisation from finitely many test cases, but as a structural consequence of the substrate's commitment to Loop, event-counting injectivity, and event-counting non-zero-successor. Induction is a quantified universal claim: $\forall n \in \mathbb{N}.\, P(0) \wedge (\forall k.\, P(k) \to P(k+1)) \to P(n)$. Observation cannot license a universal directly; no finite collection of measurements is enough. What licenses it is the structural form of the substrate's iteration: Loop applies the same step rule at every successor application, and the non-zero-successor and injectivity properties guarantee the iteration generates the substrate-naturals exactly. Anything that holds at zero and is preserved by the step rule is therefore preserved at every Loop iteration the substrate executes, regardless of the iteration count. Conservation laws and stable-mode persistence are observable consequences of this structural commitment — they are the discharge witness that the substrate *has* the iterative structure to which induction applies, not the source of the universal claim.

A residual gap remains. The substrate physically instantiates only the naturals it has actually iterated to, while mathematical induction quantifies over the type $\mathbb{N}$ including naturals never instantiated. The bootstrap closes this gap by treating $\mathbb{N}$ as the closure of the substrate's iterations under the rule rather than as a sum over instantiated cases: the type $\mathbb{N}$ is what the unbounded-iteration rule produces in principle, and induction is sound on that type because the rule acts uniformly. Whether this closure step is itself part of the substrate's primitive content (under Loop plus infinity), or a derivational closure imposed on top of the substrate's actual iterations, is a question we treat as part of the framework's metalogic-coextensivity claim of Sec. II A, not a question we resolve here.

B. Two parallel readings of arithmetic

Once the natural numbers exist, the four arithmetic operations admit two parallel substrate-physical readings. The *click-count reading* runs at the level of counted events; the *substrate-amplitude reading* runs at the level of substrate wave-patterns. Both readings are licensed by the framework's commitments and are used together in subsequent derivations. We do not derive their equivalence in this paper; we treat it as a substrate commitment, namely that the substrate's coupling-event structure is the same operational rule at the count-of-events level and at the wave-pattern level, and that the arithmetic produced at either level is therefore the same arithmetic. The corresponding bridge theorem — that the click-count ring of substrate-naturals and the amplitude-level scalar-cross-term ring are isomorphic — is stated in the framework's formal companion [11]; the present paper takes it as given. The empirical pressure on this commitment is that physics uses the same number system for both purposes; a framework that gave them distinct arithmetics would be under immediate empirical strain.

a. Click-count reading. Addition is iterated successor. To add n to m, take m and apply successor n times. A detector at click-count m that registers n more clicks ends up at $m+n$. The Loop construct iterates the successor under the Test "iteration count below n"; substrate-addition is the substrate carrying out exactly that loop. Multiplication is iterated addition: to multiply m by n, take zero and add m to it n times. A detector observed across n measurement windows, each registering m clicks, has cumulative count $m \times n$.

b. Substrate-amplitude reading. At the substrate-amplitude level, the same four operations have direct substrate-physical content as operations on substrate wave-pattern amplitudes:

- *Addition is linear superposition.* When two substrate-amplitude patterns occupy the same region of substrate position-space, the substrate's

free-propagation regime (the linear sector of its elastic response) commits that the resulting amplitude is the sum of the participating amplitudes. The substrate-physical content of addition at the amplitude level is therefore the substrate's linear superposition of two amplitude patterns.

- *Subtraction is destructive interference.* Inverse of linear superposition: removing a participating amplitude from a superposed configuration reduces the resulting amplitude by the removed amplitude's content. This is observable in the destructive interference of substrate-amplitude patterns at anti-aligned phase, where the resulting amplitude is the difference between the participating amplitudes.

- *Multiplication captures cross-term coupling.* When two substrate-amplitude patterns overlap in the substrate's nonlinear regime, the cross-term in the substrate's nonlinear response is bilinear in the participating amplitudes: doubling either input doubles the cross-term contribution; the cross-term vanishes when either input vanishes. We do not claim multiplication is the *only* mathematical structure that exhibits this bilinearity (Lie brackets, tensor products, and associative non-commutative products are all bilinear), but multiplication on $\mathbb{N}$ — commutative, associative, distributive over addition — is what the substrate's leading-order isotropic cross-term has when the participating modes are scalar amplitudes. The substrate-physical reading of multiplication at the amplitude level is therefore this bilinear cross-term in its scalar, commutative special case; more general cross-coupling structures (non-commutative, anti- symmetric, higher-rank) require the corresponding generalisations of multiplication and emerge in later sections.

- *Division is the formal inverse, with a remainder.* Inverse of cross-term coupling: separating a cross-term-coupled configuration back into its participating modes leaves a remainder consisting of the cross-term substrate-tension content that does not decompose cleanly into either participating mode individually. This remainder is what mainstream physics measures as the *binding energy* of bound configurations. We do not claim the remainder is itself the gauge-boson content (e.g. gluons in hadrons); the gauge bosons are independently characterised substrate-Lie-group connections, and the relation between the algebraic division remainder and the chromodynamic field configurations sourcing hadronic binding is non-trivial. The substrate-physical content of division at the amplitude level is the inverse cross-term operation, leaving binding-energy substrate tension as the unmodelled remainder.

Two technical caveats. First, both the click-count and substrate- amplitude readings invoke different physical regimes for $+$ and $\times$: the click-count addition uses the substrate's iteration primitive, and the amplitude addition uses the substrate's *linear* free-propagation regime, while the amplitude multiplication uses the substrate's *nonlinear* cross-term regime. The resulting arithmetic comes out the same on both readings (commutative ring with addition and multiplication), but the dynamical regimes are distinct. Second, subtraction on $\mathbb{N}$ is only partially defined ($3 - 5$ is not a natural number); the destructive-interference reading inherits this partiality, and the total subtraction operation requires the integers, constructed in Sec. IV.

The standard algebraic properties — commutativity, associativity, distributivity — follow by induction on one of the variables, using the inductive principle just established. Each property is a substrate-grounded theorem derivable mechanically from the inductive principle and the operation definitions.

IV. FROM ARITHMETIC TO THE STANDARD NUMBER SYSTEMS

A. The integers

The natural numbers count up but cannot count down past zero. To represent "removing five from a stock of three," the substrate needs negative numbers. The integer construction is the standard Grothendieck quotient: take pairs of naturals where (a, b) represents the difference $a - b$, and identify two pairs whenever their cross-sums match (i.e. $(a, b) \sim (c, d) \Leftrightarrow a + d = c + b$, an equation in $\mathbb{N}$). This construction is a derivational step on top of the substrate-naturals; what observation supplies is the existence of substrate processes whose tally values include both signs — an electric-charge accumulator with positive and negative contributions, an account balance with deposits and withdrawals, a particle counter with added and removed tallies. The discharge witness is therefore the empirical reality of opposing-tally processes; the construction-of-$\mathbb{Z}$ itself is a closure step performed in the framework's metalogic on the discharged $\mathbb{N}$. Integer operations are defined component-wise on the pair representation, using the substrate's natural-number addition and multiplication already established.

B. The rationals

The integers admit subtraction but not division. To represent "two parts in five," the substrate needs ratios. The rational construction takes pairs of integers where the second is non-zero, with (p, q) representing the fraction p/q and two pairs identified when their cross-products match. The substrate-physical content of a rational is a *ratio of two counted integer-valued substrate processes*: the number of decays in channel A over the

total number of decays in a sample (a count over a count), the count of atomic-clock cycles per second-marker, the lattice index of a stable mode within a multiplet. General real-valued ratios — branching fractions, transition frequency ratios, mixing angles — are not themselves rationals; they are real numbers approximated to arbitrary precision by Cauchy sequences of rationals, and live in the substrate-reals constructed in Sec. IV C. The discharge witness for the rationals is restricted to count-over-count quantities.

C. The reals

The rationals can represent any specific ratio but not the limit of a sequence of progressively-better rational approximations to an irrational quantity. To represent the speed of light, an atomic transition frequency, or any other measured physical quantity at arbitrary precision, the substrate needs the real numbers. The construction packages an irrational value as the equivalence class of all the Cauchy sequences of rationals that approximate it.

Two qualifications are essential here. First, observation supplies the rational *members* of any such sequence (each refined measurement is a rational with finite precision); it does not, on its own, supply the closure-to-limit step. The substrate-real is the *equivalence class*, and packaging that class as a single object is a derivational closure step performed on top of the substrate-rationals, licensed by the substrate's commitment to Loop (every Cauchy sequence is generated by an unbounded iteration) plus extensionality (two iterations producing the same limit-witnessing sequence are the same closure object). Second, the real numbers are uncountable while observations are countable: the construction does not claim every real has been measured, only that every measured quantity that admits indefinite refinement is representable as some Cauchy class.

The substrate-reals are complete by construction: any Cauchy sequence of substrate-reals has a substrate-real limit, since equivalence classes of Cauchy sequences are themselves closed under the same equivalence. Completeness is what makes substrate-reals the natural setting for limits, derivatives, integrals, and all of analysis; every measured physical quantity has, to date, been representable within the substrate-reals.

D. Sets

A substrate-set is a collection of substrate configurations sharing a defining predicate, where the predicate ranges over a previously fixed substrate-type. Every species, every catalogue, every taxonomy in physics is a substrate-set: the collection of electrons, the collection of photons, the collection of stable nuclei. Substrate-set is what substrate-type-formation produces.

Two structural restrictions are inherited from the substrate's type-formation primitive and exclude unrestricted comprehension. (i) The defining predicate must range over a previously committed substrate-type, not over the universe of all substrate-objects, so predicates of the form "x is a substrate-set that does not contain itself" are not well-formed in the type hierarchy — the substrate's type levels are stratified, and Russell-style self-reference does not type-check. (ii) The collection of all substrate-sets at one level is itself a substrate-type at the next level, never a substrate-set within the same level; the substrate-types form a cumulative hierarchy of the same shape as the type-theoretic universes used in modern formal foundations. The discharge for both restrictions is that no observed substrate-set in physics has ever required impredicative self-reference.

Set operations — subset, union, intersection, complement — combine sets at a fixed type level into new sets at the same level using the substrate's And/Or/Not-coupling rules applied to defining predicates. Physics taxonomies routinely combine sets this way: hadrons $\cup$ leptons = fermions; charged $\cap$ stable = stable charged species; complement of charged within all = neutral.

E. Topologies and continuity

A substrate-topology on a set X is a collection of "open" subsets of X that contains $\emptyset$ and X itself, is closed under arbitrary unions, and is closed under finite intersections. The notion of an open neighbourhood around a substrate configuration is observable: a measurement-precision band around a measured value, a position-uncertainty region around a localised mode, a frequency-window around a transition centre. The collection of all such observable open neighbourhoods, closed under combining and overlapping, is a substrate-topology. Two qualifications: (i) real laboratory measurements furnish only finitely many concrete neighbourhoods at any time, while a topology on $\mathbb{R}^n$ has uncountably many open sets, so the discharge here covers the *closure rules* of the topology (which laboratory practice manifestly satisfies) rather than the totality of open sets; (ii) the closure to "arbitrary unions, finite intersections" is a derivational step on top of the observed neighbourhood structure, licensed by the same closure-under-iteration rule that produces the real numbers.

A substrate-continuous map between two topological spaces preserves the open-neighbourhood structure: small input changes produce small output changes. Every measurement system is substrate-continuous; conservation laws produce substrate-continuous maps from initial states to final states. Continuity is what makes physical predictions possible. If it failed, no input precision could guarantee any output precision.

V. FROM TOPOLOGY TO GEOMETRY

A. Smooth manifolds

Topology is enough for "near-enough" comparisons but not for "rate of change" comparisons. The substrate needs smooth structure so that derivatives, accelerations, and curvature have well-defined values. A substrate-manifold is what the substrate produces when its observable open-neighbourhood structure is locally homeomorphic to a Euclidean space. The substrate has a smooth structure if every substrate configuration has a neighbourhood equipped with a chart — a homeomorphism to an open subset of $\mathbb{R}^n$ — such that on any overlap of two charts the transition map between their $\mathbb{R}^n$-images is infinitely differentiable (C^∞). The continuity of charts is what topology gives; the C^∞ compatibility on overlaps is the substrate's smoothness commitment, and is the additional structure that distinguishes a smooth manifold from a topological one.

The substrate-physical process: the substrate's elastic response is observed to be locally smooth. Small displacements produce smoothly varying restoring tensions, not discontinuous jumps. The substrate's standing-wave configurations propagate along smooth trajectories in space and time; particle worldlines are differentiable except at scattering events where new coupling events occur. The cosmic microwave background temperature field is smooth at every angular scale we can resolve. The substrate's smoothness is not a postulate; it is observed in every measurement that requires rate-of-change content.

A substrate-tangent-vector at a configuration is what falls out when the substrate's smoothness is differentiated: the directional rate-of-change of any substrate-observable property along an infinitesimal substrate-displacement. Every measurement of "how this changes when you move that way" — drift velocities, pressure gradients, field strengths, gauge-potential gradients — is the substrate producing a tangent vector.

B. Tensor fields

A substrate-tensor at a configuration is what the substrate produces when it couples several tangent vectors and dual covectors in a multilinear way. The substrate's stress-energy tensor at a location is rank-2; the Riemann-curvature tensor is rank-4. Every observed multi-directional substrate-physical quantity — a stress, a strain, a moment of inertia, an electromagnetic field, a metric — is a substrate-tensor.

The substrate's elastic response naturally couples multiple directions: pulling in one direction produces stress in several others, depending on local geometry. The multi-direction coupling is observed to be multilinear: doubling the input strain in any one direction doubles the output stress in each output direction; adding two input strains adds the two output stress responses. Without multilinearity, the superposition principle that lets us decompose substrate fields into independent modes would fail. Tensor calculus is the formal capture of this observed multilinear coupling.

C. Differential forms and exterior calculus

A substrate-differential-form is a substrate-tensor that is *antisymmetric* under exchange of its tangent-vector slots. The antisymmetry comes from a substrate-physical observation: oriented quantities — area swept, volume occupied, flux through a surface — change sign under orientation reversal. Every observed substrate-orientation phenomenon (magnetic flux through a loop, vortex circulation, fluid-volume flow, neutrino helicity, weak-interaction parity violation) produces a differential form.

The substrate's exterior derivative d produces, from any oriented-displacement form, a higher-rank oriented-displacement form measuring how the lower form fails to close. Two related identities appear here, and they are not the same statement. The algebraic identity $d^2\omega = 0$ for every form ω follows from the exterior derivative's definition together with the equality of mixed partial derivatives $\partial_i\partial_j = \partial_j\partial_i$; the topological identity $\partial\partial M = \emptyset$ (the boundary of a boundary is empty) is a property of the boundary operator on chains. Stokes' theorem $\int_{\partial\Omega}\omega = \int_\Omega d\omega$ is the duality that connects them: applying the algebraic identity inside the integral matches the topological identity on the integration domain. The substrate-physical content is the existence of *both* identities: oriented displacements compose without remainder ($d^2 = 0$ on forms), and oriented regions have boundaries whose boundaries vanish ($\partial^2 = \emptyset$ on chains). Conservation laws are theorems of this paired commitment.

As a worked example, the homogeneous Maxwell equations $dF = 0$ are exactly the algebraic identity $d^2 = 0$ applied to a gauge potential whenever $F = dA$ holds. On a topologically trivial region this reduces to "no magnetic monopoles plus Faraday's law"; on a region with non-trivial second cohomology H^2, F may be closed but not exact, and $H^2 \neq 0$ is exactly the condition that admits magnetic monopoles. The no-monopole content of Maxwell's equations is therefore equivalent to the cohomological vanishing $H^2 = 0$ in the substrate's gauge bundle, and our writing $F = dA$ is the assertion that the substrate's electromagnetic gauge bundle is, on the relevant region, trivial enough for the potential A to exist globally.

The substrate's Stokes' theorem identifies the integral of a form over a region with the integral of its exterior derivative over the boundary: $\int_{\partial\Omega}\omega = \int_\Omega d\omega$. This is the substrate's flux conservation in its most general form. Every observed conservation law (charge, energy, momentum, angular momentum, baryon number, lepton number) is Stokes' theorem applied to the appropriate closed differential form.

D. Vector calculus and fluid dynamics

Vector calculus is the special case of exterior calculus on $\mathbb{R}^3$. The gradient is the exterior derivative on 0-forms (scalars), giving a 1-form identified with a vector field via the substrate-metric. The curl is the exterior derivative on 1-forms, giving a 2-form, identified with a vector field via the Hodge star $\star$. The divergence is the exterior derivative on 2-forms, giving a 3-form, identified with a scalar via $\star$. All three are special cases of the single exterior-derivative operator d on the corresponding form-degree. The substrate-physical content of each is direct: the gradient is the direction of fastest substrate-tension increase; the divergence is the net outflow of substrate-flow through an infinitesimal region; the curl is the rotation of substrate-flow around an infinitesimal loop. Every continuity equation, every Euler equation, every Navier–Stokes equation expresses, at the appropriate scale, the substrate's local commitment to "no spontaneous creation or destruction of conserved quantities."

VI. FROM GEOMETRY TO SYMMETRY

A. Lie groups

A substrate-Lie-group is what the substrate produces when a continuous family of its symmetries is itself smooth. Every continuous symmetry the substrate exhibits — spatial rotation, spatial translation, time translation, gauge phase rotation, isospin rotation, Lorentz boost — is a Lie-group element, and the family of all such elements forms a substrate-manifold G on which both the group multiplication $G \times G \to G$ and the inversion $G \to G$ are smooth maps.

The substrate-physical process: substrate standing-wave configurations are observed to remain physically equivalent under continuous deformations of certain kinds. Rotating the laboratory frame does not change atomic spectra; performing a global gauge phase rotation on a wavefunction does not change probability densities; an isospin rotation does not change strong-interaction cross-sections within an isospin multiplet. Each is a substrate-Lie-group acting on substrate configurations.

B. Lie algebras

A substrate-Lie-algebra is what the substrate produces when a Lie group's smoothness is differentiated: the tangent space at the identity element. Lie-algebra elements are infinitesimal substrate-symmetry generators; the Lie bracket is what falls out when two infinitesimal symmetries are composed and the order of composition matters. Every observed symmetry-non-commutativity phenomenon — Thomas precession in special relativity, the $SU(2)$ commutation of angular-momentum components, the commutator structure of strong- interaction colour generators — is a substrate-Lie-bracket.

The substrate's exponential map $\exp : \mathfrak{g} \to G$ from Lie algebra to Lie group is the substrate's commitment to integrating an infinitesimal symmetry to a finite one. Every observed continuous-symmetry-generation phenomenon — a magnetic field precessing a spin, a Lorentz boost combining time and space coordinates, a quark-mass-matrix diagonalisation rotating from flavour to mass eigenstates — is the exponential map applied.

C. Root systems

A substrate-root-system is what the substrate produces when a Lie algebra's adjoint action on itself is decomposed into Cartan- subalgebra eigen-directions: the non-zero *roots* $\alpha \in \mathfrak{h}^*$ are the eigenvalues of the Cartan action on $\mathfrak{g}$, and they form a substrate-set with a specific reflection-symmetric geometric structure. The same Cartan action on any representation V decomposes V into weight spaces; the *weights* of V are the corresponding eigenvalues. Roots are the weights of the adjoint representation specifically.

The substrate's standing-wave catalogue exhibits quantised charges on observed states: integer multiples of $e/3$ for electric charge (quarks at $\pm e/3, \pm 2e/3$ and integer-charge composites at multiples of e), half-integer or integer spin, the three states of the $SU(3)$ fundamental representation for colour, two states of the $SU(2)$ weak-isospin doublet, and so on. These quantum numbers are *weights* of the substrate-Lie-algebra acting on the representations carried by physical states, not roots *per se*; only the gauge bosons (which transform in the adjoint) carry roots as their charges. The quantisation arises because both roots and weights are constrained to a discrete lattice by the algebra's integrality conditions, and the substrate's observable charges populate this lattice. The three matter generations are a separate datum — a multiplicity of replicas of the same charged-fermion representation, not a weight value — fixed in the framework by the Hodge structure of the Schoen quotient [12] rather than by the root/weight lattice.

The substrate's roots satisfy a constraint that is a *theorem* of the substrate-Lie-algebra structure, not an empirical observation: the integrality condition $\langle \alpha, \beta^\vee \rangle \langle \beta, \alpha^\vee \rangle \in \{0, 1, 2, 3\}$ on any pair of non-proportional roots forces the angle between them into the discrete set $\{30°, 45°, 60°, 90°, 120°, 135°, 150°\}$. This is the Killing–Cartan crystallographic constraint [10], derivable from the Lie-algebra primitives the substrate has already committed to (commutator structure plus the Cartan-subalgebra eigen-decomposition). What is empirical is the *instantiation*: the substrate's stable mode lattices are observed to occupy positions consistent with one of the finite list of root systems the theorem permits, rather than violating the constraint. The periodic table, the meson–baryon multiplet structure, and the Standard

Model gauge-group decompositions are the empirical witnesses; the angle constraint itself is what the theorem requires of any root system.

D. Coxeter and Weyl groups

A substrate-Coxeter-group is the discrete reflection-symmetry group of the substrate's root system. The substrate's standing-wave catalogue exhibits exact mirror symmetries: every substrate-root has a negative counterpart; reflection across the hyperplane perpendicular to any substrate-root is a symmetry of the substrate's root system. The group generated by all such reflections is the substrate's Weyl group, and its abstract presentation in terms of generators and order relations is the substrate's Coxeter group. The Coxeter element — the product of all simple reflections, taken in any order — has order h as a group element (the *Coxeter number*); its eigenvalues on the Cartan subalgebra are the r complex numbers $e^{2\pi i\, m_k/h}$ where the integers $\{m_1, \ldots, m_r\}$ are the *exponents* of the algebra and r is the rank, with $|\Phi| = h \cdot r$ the size of the root system [10].

E. The empirical anchor: E_8 in $CoNb_2O_6$

The construction so far is structural. It says that a substrate-Lie- algebra exists with a specific root system and a specific Coxeter group. It does not, in itself, say which specific Lie algebra the substrate selects. That selection is empirical, and the selection has been observed.

Coldea *et al.* [7] measured the spectrum of the quasi-one-dimensional Ising ferromagnet $CoNb_2O_6$ near its quantum critical point. The lowest two excitation masses were found to satisfy $m_2/m_1 = \varphi = (1+\sqrt{5})/2$ to part-in-10^3 precision — the golden ratio. This is the leading mass ratio of the Zamolodchikov E_8 spectrum [8]: eight integrable particle masses indexed by the nodes of the E_8 Dynkin diagram, with the mass values fixed (up to overall scale) by the Perron–Frobenius eigenvector of the E_8 Cartan matrix.

The empirical content: E_8 root-system structure is observable in the laboratory at the level of mass ratios in a system that knows nothing about high-energy physics. This is the framework's evidence that the substrate's mode catalogue selects *at least one* copy of the E_8 root system rather than a generic Lie algebra. The further identification of $E_8 \times E_8$ as the substrate's full symmetry algebra is a separate step: the doubling has its own motivation in heterotic-string anomaly cancellation and modular invariance [11, 12], and the $CoNb_2O_6$ observation does not, by itself, force the doubling. We treat the single-E_8 part as discharged by Coldea and the doubling as a separate framework commitment with its own empirical inputs (the cancellation of gauge anomalies in the Standard Model, the visible-vs-dark sector splitting). The bootstrap chain in the present paper runs only as far as "substrate-Lie- algebra with one or more E_8 factors"; the specific selection of the doubled heterotic algebra is established in the companion work.

F. The McKay correspondence and icosahedral seeding

The McKay correspondence [9] establishes a bijection between finite subgroups of $SU(2)$ and the simply-laced ADE Dynkin diagrams via the irreducible representations on resolution divisors of the corresponding du Val singularities. The binary icosahedral group is the McKay dual of E_8. The McKay correspondence is, in itself, a representation-theoretic statement about resolutions of $\mathbb{C}^2/G$ singularities; it does not, on its own, predict mass ratios. The Zamolodchikov E_8 mass spectrum measured by Coldea *et al.* arises from the integrable structure of the perturbed Ising CFT, not from the McKay correspondence directly. The framework's claim is the more modest statement that the McKay-dual binary icosahedral group — and in particular its order-5 rotational substructure — is structurally available as a discrete symmetry whenever the substrate's E_8 acts; the framework then *selects* this order-5 substructure as the harmonic seed of its mass-spectrum construction in the companion paper [12]. The selection is a framework commitment, not a McKay theorem; the empirical content that discharges it is the agreement of the resulting φ-chain predictions with measured fermion masses and with the $CoNb_2O_6$ golden-ratio mass ratio at the appropriate level of the Coxeter spectrum.

VII. THE DISSOLUTION OF WIGNER'S PUZZLE

We can now state the resolution. The argument is short. By *structural identity* we mean specifically: the formal description licensed by the substrate's primitive content, when restricted to the slice of mathematics enumerated in Sec. II A (elementary number theory through Coxeter classification), produces the same objects, up to bridge-theorem isomorphism, as the standard-mathematics objects of those theories. This is an isomorphism claim between two presentations, not a metaphysical identification of mathematics with physics simpliciter.

1. Every primitive of *the mathematics that physics uses*, from the natural numbers through Coxeter groups, has been derived above from the substrate's primitive content. Each derivation is paired with an observable discharge.

2. The substrate's primitive content is, by the framework's implication-graph construction, traceable back to a single ontological root: the substrate exists.

3. Within that scope, mathematics is what falls out of the substrate's primitive content when that content is described formally; the bootstrap-derived objects and the standard-mathematics objects are isomorphic by the bridge theorems (Sec. II A). We do not extend this claim to mathematics outside the scope.

4. Applying that slice of mathematics to physics is therefore not a contingent mapping between two distinct domains. The formal description and the substrate-physical content it describes are the same object up to bridge-theorem isomorphism, and the description's correctness on the physics is a consequence of that isomorphism rather than a separately discovered fit.

The "unreasonable effectiveness" of mathematics in the natural sciences is, on this reading, the consequence of a structural identity rather than a coincidence between two independent objects. The mathematics that appears in successful physics is the formal description of substrate-physical content; the physics that mathematics describes successfully is the substrate's actual behaviour. The empirical content of any physical theory — a measurement of an oscillation frequency, a charge, a decay width — is a substrate process running, and the mathematical description of that process is the formal capture of the same substrate content. The puzzle dissolves because the assumption it required — mathematics on one side, physics on the other, with a contingent mapping in between — is not a feature of the framework. The "mapping" is the formal description, and the description's correctness on the physics is bound to follow because the description is, by construction, that content's formalisation.

This explains, without further explanation, why the parts of mathematics that are most useful to physics are the parts that formalise substrate-physical content directly. Linear superposition is useful because the substrate has a linear regime. Differential forms are useful because the substrate's coupling events are oriented. Lie groups are useful because the substrate has continuous symmetries. Root systems are useful because the substrate's mode catalogue is quantised. Each is the formal description of an observed substrate-physical phenomenon, and the formal description is bound to be effective on the phenomenon it formalises.

It also explains, structurally, the cases that surprised Wigner. Complex numbers are not a calculational convenience that surprised us by being irreducible to quantum mechanics. The substrate's wave-pattern amplitudes carry an oriented two-dimensional structure (magnitude and phase) under which composition acts by rotation: multiplication by $e^{i\theta}$ rotates a substrate amplitude by angle θ in the amplitude plane, and in particular multiplication by i is the quarter-turn (so that $i^2 = -1$ is the half-turn that takes an amplitude to its negative, and $i^4 = 1$ returns it to identity). The complex numbers are the smallest field that supports this oriented rotational composition algebraically; their appearance wherever the substrate's amplitude phase matters (interference, gauge phase, Bloch states, the Schrödinger equation) is structural rather than coincidental. Hilbert spaces are similarly not an abstraction that surprised us by being the right home for quantum states. The substrate's standing-wave amplitudes form a vector space under linear superposition (which the substrate already commits to in its free-propagation regime), and the choice of inner product reflects the substrate's amplitude pairing under which $|\psi|^2$-type integrals — conserved quantities, transition probabilities, mode overlaps — are the appropriate physical observables. The completion of this vector space under the corresponding norm is a Hilbert space; the L^2 choice of pairing is what makes the substrate's continuity equation $\partial_t|\psi|^2 + \nabla \cdot \mathbf{j} = 0$ consistent with the unitary linear dynamics of free propagation. We do not claim Hilbert-space completeness or the L^2 choice are *forced* from substrate primitives alone in this paper; we claim the framework's free-propagation, superposition, and conserved-amplitude commitments together select Hilbert-space structure as their natural formal capture, which is why Hilbert spaces turn out to be where quantum mechanics lives.

VIII. LIMITS OF THE PRESENT CONSTRUCTION

We state the limits plainly. The bootstrap chain above is structural and observationally anchored at every step, but several pieces are provisional.

- *Selection of the specific algebra.* The chain produces Lie algebras and root systems in general but does not, in itself, force $E_8 \times E_8$ specifically. The E_8 identification is empirical, anchored in the Zamolodchikov spectrum measured in $CoNb_2O_6$ [7, 8]; the doubling to $E_8 \times E_8$ has separate empirical motivation in heterotic-string anomaly cancellation and the visible-vs-dark sector split of the companion paper's [12] mass spectrum. Upgrading the algebra selection from empirical to structurally forced would require a substrate-physical projection onto the heterotic moduli space — the same open work flagged in the companion paper.

- *Discharge completeness.* Most of the substrate-mathematical hypotheses admit immediate discharge from the substrate's coupling-event definition or direct observation. Two — abstraction congruence and the strongest form of type instantiation — are provisionally discharged pending the substrate's standing-wave-mode catalogue, which is supplied elsewhere in the framework [11] but is not part of the present bootstrap. The provisional discharges are sufficient for everything in the present paper.

- *Construction-and-bridge equivalence.* Each substrate-derived object has both an existence-form definition (anchored in observation) and a constructive form (mirroring standard mathematics). The two presentations are connected by bridge theorems showing they describe the same object. The bridge theorems are stated and machine-verified throughout the framework's formal companion [11]; the present paper takes them as given and refers the reader to the formal record.

- *Higher mathematics.* The bootstrap above runs from the natural numbers through Coxeter groups. Higher structures — algebraic geometry, sheaves, derived categories, ∞-toposes — have not been bootstrapped here. The framework's expectation is that they admit substrate-physical readings of the same character (sheaves as substrate-mode patterns over base manifolds, derived categories as the substrate's homological structure, etc.), but the present paper does not work that out.

- *Quantum mechanics, general relativity, and the SM.* The application of the bootstrap to specific physics frameworks is summarised in the companion paper [12] and developed in detail in [11]. The Standard Model parameter spectrum, Newton's constant, and a unified neutrino–dark-matter tower are derived from a single anchor (the electron rest mass) in that work; the present paper provides the foundational mathematics on which those derivations rest.

IX. CONCLUSION

Wigner's question — why is mathematics so unreasonably effective in physics? — admits a structural answer once the assumption underlying it is dropped. The assumption was that mathematics and physics are two different things and that the success of mathematics in physics is a coincidence requiring explanation. We have argued that, in a substrate-physical framework with a single ontological root (the substrate exists), every primitive of *the mathematics physics uses* — the slice enumerated in Sec. II A — is derivable from the substrate's primitive content and discharged by an observable phenomenon. Mathematics is not a separate body of structure that happens to fit physics; it is the formal description of the substrate's primitive content. Its effectiveness in physics is structural: the mathematics that appears in physics is, by construction, the formalisation of the substrate-physical content that the physics measures.

The construction is observational at every step. Each substrate-mathematical object listed has an anchoring phenomenon. Each step is paired with a discharge obligation that has been either directly satisfied or is in progress. The framework's implication graph terminates, transitively, at the single root commitment that the substrate exists. We do not claim the construction is complete — the open items in Section VIII flag what remains — but the structure is in place, and the empirical anchor for the specific E_8 algebra has been observed in the laboratory.

If correct, the consequence is that for the slice of mathematics that physics actually uses, "unreasonable effectiveness" is a category mistake: that slice and the physics it describes are not two contingently related domains but two levels of formalisation of the same substrate-physical content. We do not extend the claim beyond that slice. Mathematics whose objects have no observed substrate-physical instantiation — large cardinals, certain choice-dependent constructions, exotic foundations, and so on (Sec. II A) — is consistent with the present framework but is not claimed to be the substrate's formalisation. The dissolution of Wigner's puzzle is therefore narrow and load-bearing rather than universal: the parts of mathematics that fit physics fit because they describe the substrate, and the parts that do not fit are unaffected by the present argument.

ACKNOWLEDGMENTS

The framework summarised here was developed across many threads of research collected in [11]; the bootstrap material draws on the foundational mathematics treatment in that work. The companion paper [12] carries out the quantitative-physics applications that the present paper grounds.

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